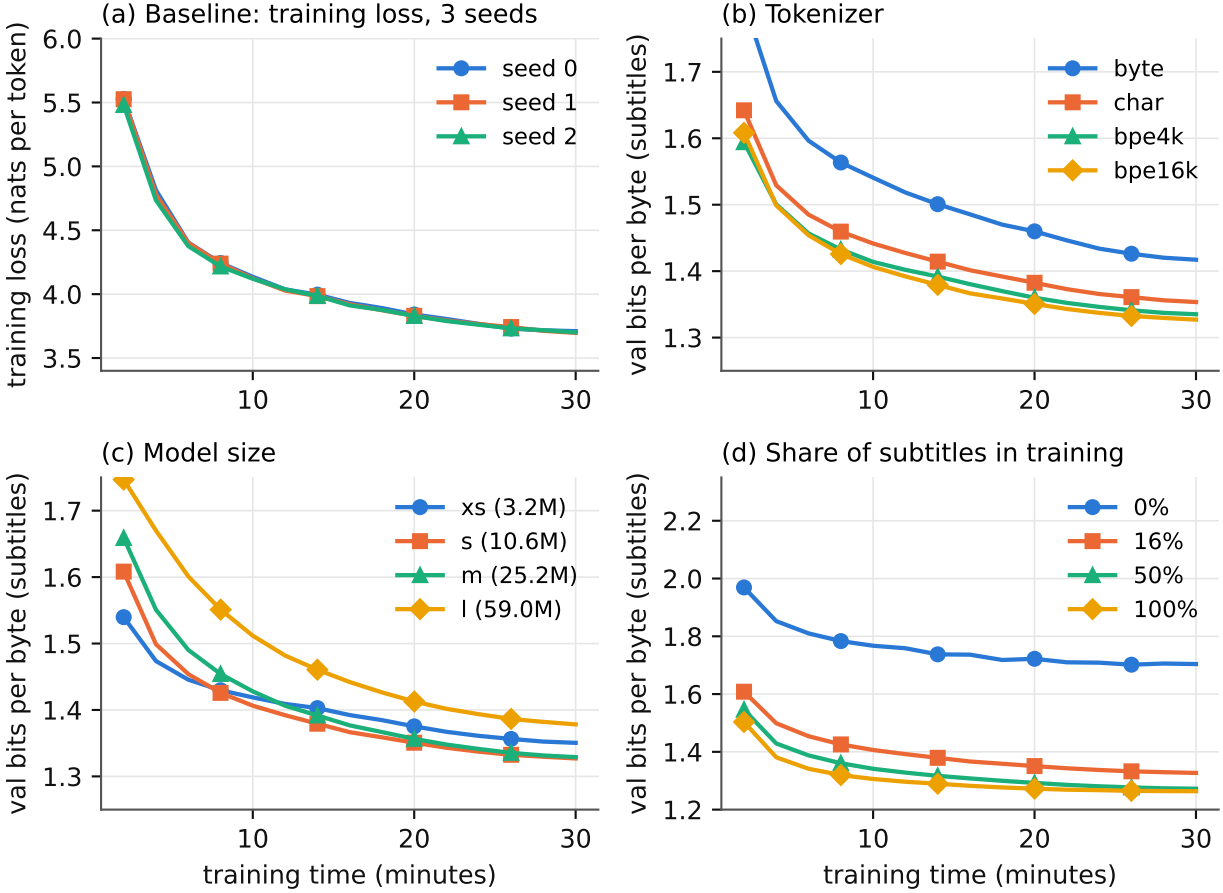
(a) Baseline: training loss, 3 seeds
training loss (nats per token)
6.0
5.5
5.0
4.5
4.0
3.5
10
20
30
seed 0
seed 1
seed 2
(b) Tokenizer
val bits per byte (subtitles)
1.7
1.6
1.5
1.4
1.3
10
20
30
byte
char
bpe4k
bpe16k
(c) Model size
val bits per byte (subtitles)
1.7
1.6
1.5
1.4
1.3
10
20
30
training time (minutes)
xs (3.2M)
s (10.6M)
m (25.2M)
l (59.0M)
(d) Share of subtitles in training
val bits per byte (subtitles)
2.2
2.0
1.8
1.6
1.4
1.2
10
20
30
training time (minutes)
0%
16%
50%
100%